
Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

[Presenter name]

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THE PROBLEM

Monetary policy stalls when money is hoarded and rates cannot fall below zero

MONEY IS NOT SPENT

€8.8tn

Eurosystem balance sheet at its peak, June 2022

yet inflation averaged 0.3% in 2014–16:
new money was held, not spent

RATES CANNOT FALL

8 years

with the deposit facility rate at or below zero, 2014–22

floor at -0.50% : below that, savers switch to cash, which pays zero

$$M \downarrow V = PY$$

$$i \geq -c_{\text{cash}}$$

A TEMPTING FIX

Vouchers with a deadline get money spent, and a programmable digital euro could become one

CHINA, DIGITAL COUPONS, 2020

¥3

extra spending per ¥1 of subsidy

FRANCE, RANDOMISED PREPAID CARDS

61% vs 23%

one-month MPC with vs without a three-week expiry

A digital euro is programmable. Could money itself carry a deadline?

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024).

THE FLAW

But an expiry date pushes out more spending, ends in a fire sale and splits money into vintages

MODEL

A euro is worth V to its holder. A good purchase is worth 1, a fire sale $\phi < 1$. Searching at speed κ costs $\kappa^2/2a$, so

$$\kappa^* = a(1 - V)$$

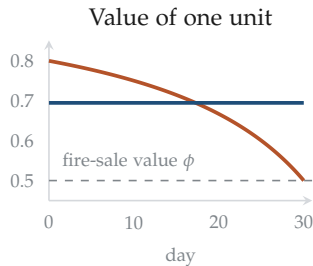
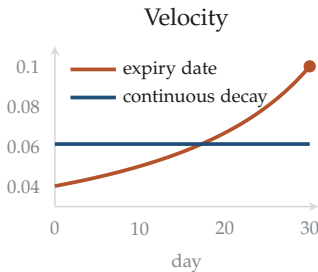
84% vs 73%

useful spending within 30 days

16%

of issuance dumped on the last day

Both schemes calibrated to the same average velocity over 30 days; $a = 0.2$, $\phi = 0.5$, $r = 4\%$.



0.80 $\rightarrow$ 0.50

value of one euro, by its age

Continuous decay keeps every euro equal: it is simply a negative rate that reaches every wallet

	Expiry	Decay
Fire sale at the deadline	3–55%	none
Value of one unit	by age	equal
One euro = one euro	no	yes
Applies to all money	no	yes
Is it an interest rate?	no	$i = -\lambda$

VELOCITY UNDER DECAY

$$\kappa^* \approx \sqrt{2a(r + \lambda)}$$

the square-root rule of Baumol–Tobin money demand

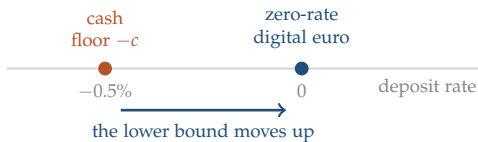
Decay λ	1%	2%	3%
Velocity	+12%	+22%	+32%

It unblocks both channels: money moves faster, and the rate reaches retail balances.

Yet the digital euro is set at zero, which makes it cash without storage costs and raises the lower bound

"The digital euro shall not bear interest."

Draft regulation on the digital euro, Art. 16(8)



Banks cannot pay less than the best central-bank alternative:

$$r_D \geq \max_j \{ i_j - c_j \}$$

Cash: $0 - c < 0$. Digital euro:
 $0 - 0 = 0$.

The lower bound and disintermediation are one problem: a central-bank rival to deposits.

A hard holding limit stops runs in a crisis, but blocks ordinary users in normal times

NORMAL TIMES: MONEY FOR PAYMENTS

18%

of people want to hold more than €3,000

A cap blocks them. A price would only make them pay.

⇒ price is enough

CRISIS: FLIGHT TO SAFETY

€4.7tn

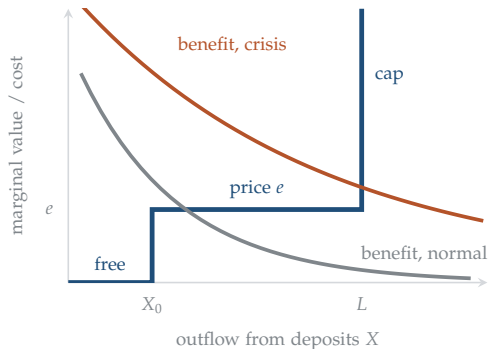
deposit outflow without a cap (€0.65tn with €3,000)

A rate gap of 2% means nothing next to the fear of losing a deposit.

⇒ only quantity works

Survey demand: Lambert et al. (ECB WP 2980, 2024). Outflows: own calibration to ECB technical data on holding limits (Oct 2025).

When costs have a cliff and benefits do not, price and quantity together beat either alone



WEITZMAN (1974)

Advantage of prices over quantities:

$$\Delta = \frac{\sigma^2 (b - c)}{2 b^2}$$

b : slope of benefit, c : slope of cost

- Flat cost, steep benefit: **price**
- Vertical cost at L : **quantity**
- Both on one curve: **tiered rate with a cap** (Roberts & Spence 1976)

The optimal digital euro has three numbers: a free tier, a Pigouvian rate above it, and a cap

FREE TIER M_1

€510

$$N \mathbb{E}[\min(t, M_1)] = X_0$$

payments use only what banks

absorb for free (€127bn)

RATE ABOVE THE TIER

−2.4%

$$-e = -(\text{MRO} - r_D)$$

holders pay the bank's cost of

replacing the deposit

CAP $\bar{M}$

€3,000

$$N \mathbb{E}[\min(d, \bar{M})] = \theta D$$

crisis outflow stays within bank

liquidity tolerance

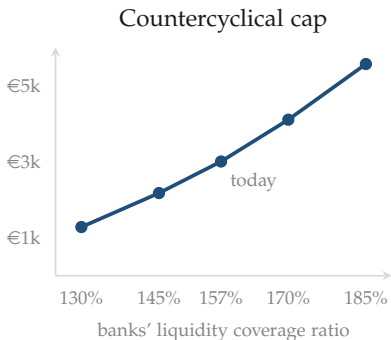
	Users blocked	Outflow if rates < 0	Outflow in a crisis
Cap only (0%, €3,000)	18.5%	€648bn	€648bn
Price only (tier, −2.4%)	0%	€158bn	€4,716bn
Hybrid (tier, −2.4%, cap)	7.3%	€158bn	€648bn

Calibrated to ECB WP 2980 survey distributions and ECB holding-limit outflows (Oct 2025); MRO 2.65%, household overnight rate 0.26%.

CONCLUSION

The digital euro should bear interest: zero for payments, decay above, and a cap that moves with the cycle

1. Deadlines get money spent, but expiry belongs to a voucher, not to money
2. Decay keeps every euro equal and is simply a negative rate
3. A zero-rate digital euro is costless cash: it raises the lower bound
4. Free tier, Pigouvian decay, and a cap tied to bank liquidity



Backup

Expiry versus decay: derivation

Holder's problem. Per unit of balance, with decay λ :

$$(r + \lambda)V = \max_{\kappa} \left\{ \kappa(1 - V) - \frac{\kappa^2}{2a} \right\} \Rightarrow \kappa^* = a(1 - V), \quad (r + \lambda)V = \frac{a}{2}(1 - V)^2.$$

Square-root rule. With $x = 1 - V$ small, $(r + \lambda)(1 - x) = \frac{a}{2}x^2$, so $x \approx \sqrt{2(r + \lambda)/a}$ and $\kappa^* \approx \sqrt{2a(r + \lambda)}$.

Expiry. With $\tau = T - t$ left: $dV/d\tau = -rV + \frac{a}{2}(1 - V)^2$, $V(0) = \phi$. If $\frac{a}{2}(1 - \phi)^2 > r\phi$, $V > \phi$ before T , so the fire sale happens only at T . Velocity rises to $a(1 - \phi)$: finite, but the unspent balance is sold at once.

Calibration. λ matches the average expiry velocity over $[0, T]$.

T	useful (expiry)	dumped	value range	λ p.a.	useful (decay)	decayed
7 d	45%	55%	0.50–0.63	1172%	41%	15%
30 d	84%	16%	0.50–0.80	489%	73%	16%
90 d	97%	3%	0.50–0.91	161%	88%	10%

Prices versus quantities: derivation

Benefit $MB(q) = b_0 + \theta - bq$, cost $MC(q) = c_0 + cq$, shock θ with variance σ^2 ; optimum $q^*(\theta) = (b_0 + \theta - c_0)/(b + c)$.

- Quantity fixed at $\mathbb{E}q^*$: expected loss $\frac{\sigma^2}{2(b+c)}$
- Price fixed: q moves by θ/b , expected loss $\frac{\sigma^2 c^2}{2b^2(b+c)}$
- Difference: $\Delta = \frac{\sigma^2(b-c)}{2b^2}$. Prices win iff $b > c$.

Three-segment cost. $MC(X) = 0$ for $X \leq X_0$, e for $X_0 < X \leq L$, ∞ beyond L . The planner's optimum is decentralised by a zero rate up to M_1 , the rate $-e$ above it, and a cap at $\bar{M}$:

$$N \mathbb{E}[\min(t, M_1)] = X_0, \quad i_2 = -e, \quad N \mathbb{E}[\min(d, \bar{M})] = \theta D.$$

Tier-2 demand shrinks by $K = \sqrt{r_D/(r_D + e)} = 0.31$ (Baumol–Tobin).

Calibration: data and fit

Sight deposits per person	median €2,000, mean €13,100	ECB WP 2980
Desired digital-euro holdings	median €560, mean €3,220	ECB WP 2980
Crisis outflow	€156bn at €500; €699bn at €3,000	ECB, Oct 2025
Digitalisation inflow by 2034	€127bn	ECB, Oct 2025
MRO; household overnight rate	2.65%; 0.26%	ECB, Sep / Apr 2026
Aggregate LCR	156.7%	ECB, Q3 2025

Fit. Lognormals from the survey; population fitted to the two outflows: $N = 360\text{m}$ (euro area 2026: 359.6m). Model: €156bn at €500, €648bn at €3,000; mean holding under a €3,000 cap €1,086 (survey: €1,100); 18% want more than €3,000 (survey: 20%). No cap: €4.7tn (Meller & Soons 2023: almost €5tn).

Sensitivity. X_0 of €100bn / 200bn / 300bn gives M_1 of €370 / 960 / 1,840. Tolerance θ of 5% / 7.6% / 10% gives $\bar{M}$ of €1,690 / 3,000 / 4,520.

Caveat. e is the private cost of replacing deposits; with central-bank pass-through funding the social cost may be lower.

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